Unit 1

Vectors and Matrices

Lesson 1.1: Vectors in $\mathbb{R}^n$

Learning Outcomes: At the end of the lesson, the students are able to:

1. perform vector addition and scalar multiplication of vectors;
2. represent vectors geometrically;
3. establish properties of the basic vector operations;
4. determine the linear combinations of a given set of vectors.

In the physical sciences, *vectors* are quantities which possess both *magnitude* and *direction*. Some of the well-known examples of vector quantities are displacement velocity, acceleration, force, and weight.

Vectors in linear algebra are the abstraction of the physical vectors. They are quantities that act on a given space with a specified coordinate system. As we shall see, these vectors are not too different from the ones we have learned in physical science courses.

Basic Vector Operations

Definition 1

Let n be a positive integer. A **real n-vector** is an ordered n-tuple of real numbers usually expressed as

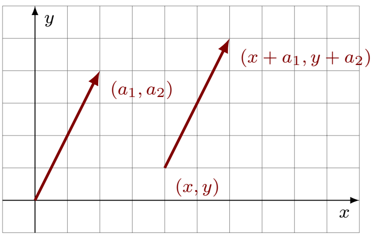
$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

where $x_i \in \mathbb{R}$, for $i = 1, 2, \ldots, n$. The element x_i is called the **ith component**. The set of all real n-vectors is denoted by $\mathbb{R}^n$ and two n-vectors are **equal** if they are equal component-wise.

When $n = 2$, the elements of $\mathbb{R}^2$ are called **plane vectors**. A plane vector

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

may be represented geometrically as an arrow whose *initial point* is at any point (x, y) in the cartesian plane such that its *terminal point* is at $(x + a_1, y + a_2)$.



The figure on the left shows two representations of the vector $\vec{a}$. The one whose initial point is at the origin is said to be in *standard position*. The vector $\vec{a}$ may be thought as a translation of the plane moving every point a_1 units in the positive x-direction and a_2 units in the positive y-direction.

Definition 2: Vector Addition

If $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$, we define the **sum**

$$x + y = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}.$$

Definition 3: Scalar Multiplication

If $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $c \in \mathbb{R}$ is a *scalar*, we define $c\vec{x} = c \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{bmatrix}$.

The vector in $\mathbb{R}^n$ whose components are all 0 is called the **zero vector** denoted by $\vec{0}$. The **negative** of the vector $\vec{x}$ is defined as

$$-\vec{x} = \begin{bmatrix} -x_1 \\ -x_2 \\ \vdots \\ -x_n \end{bmatrix}.$$

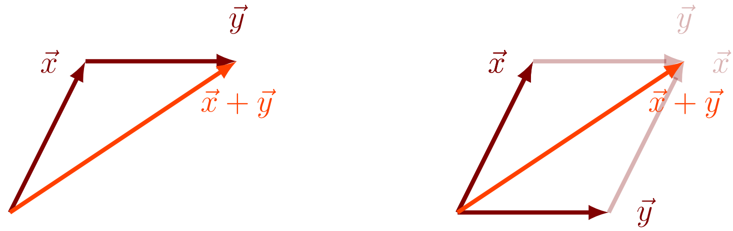
We also define the **difference** $\vec{x} - \vec{y} = \vec{x} + (-\vec{y})$.

Example 4. Given that $\vec{x} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $\vec{z} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$. Calculate $2\vec{x} + 3\vec{y} - \vec{z}$.

Solution.

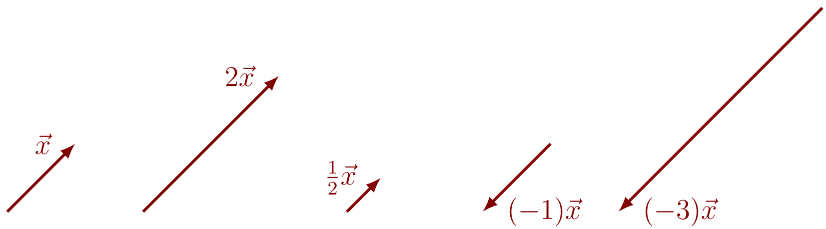
$$\begin{aligned} 2\vec{x} + 3\vec{y} + \vec{z} &= 2\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + 3\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} + \begin{bmatrix} -3 \\ 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -4 \\ 0 \\ 4 \end{bmatrix}. \end{aligned}$$

In the case when $n = 2$ or 3, one may think of sum of $\vec{x} + \vec{y}$ as the net effect when a translation by $\vec{x}$ is composed with a translation by $\vec{y}$. Geometrically, one may choose arrow representations of $\vec{x}$ and $\vec{y}$ so that the terminal point of $\vec{x}$ is the initial point of $\vec{y}$. Consequently, the sum $\vec{x} + \vec{y}$ may be represented by the arrow whose initial point is that of $\vec{x}$ and whose terminal point is that of $\vec{y}$. Equivalently, when the representations of $\vec{x}$ and $\vec{y}$ are chosen in standard positions, the sum $\vec{x} + \vec{y}$ in standard position is the arrow the lies in the parallelogram formed by $\vec{x}$ and $\vec{y}$.



This manner of representing the sum of two vectors is also known as the **parallelogram rule**.

If $\vec{x}$ is a nonzero vector and $c \in \mathbb{R}$ is a scalar, the scalar multiple $c\vec{x}$ is the vector whose length is $|c|$ times the length of $\vec{x}$ and whose direction is the same as $\vec{x}$ when $c > 0$ and opposite to $\vec{x}$ when $c = 0$.



One observes that $(-1)\vec{x} = -\vec{x}$ and that $0\vec{x} = c\,\vec{0} = \vec{0}$ for any vector $\vec{x}$ and any scalar c.

Theorem 5: Properties of Vector Operations

Let $\vec{x}, \vec{y}, \vec{z}$ be vectors in $\mathbb{R}^n$ and c, d be scalars. Then

1. $\vec{x} + \vec{y} = \vec{y} + \vec{x}$.
2. $\vec{x} + (\vec{y} + \vec{z}) = (\vec{x} + \vec{y}) + \vec{z}$.
3. $\vec{x} + \vec{0} = \vec{x}$.
4. $\vec{x} + (-\vec{x}) = \vec{0}$.
5. $c(\vec{x} + \vec{y}) = c\vec{x} + c\vec{y}$.
6. $(c + d)\vec{x} = c\vec{x} + d\vec{x}$.
7. $(cd)\vec{x} = c(d\vec{x})$.
8. $1\vec{x} = \vec{x}$.

We shall write a proof for (5) and leave the other proofs as exercises.

Proof of (5). Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$ and let c be a scalar. Then

$$\begin{aligned} c(\vec{x}+\vec{y}) &= c\left(\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}\right) = c\begin{bmatrix} x_1+y_1 \\ x_2+y_2 \\ \vdots \\ x_n+y_n \end{bmatrix} \\ &= \begin{bmatrix} c(x_1+y_1) \\ c(x_2+y_2) \\ \vdots \\ c(x_n+y_n) \end{bmatrix} = \begin{bmatrix} cx_1+cy_1 \\ cx_2+cy_2 \\ \vdots \\ cx_n+cy_n \end{bmatrix} \\ &= \begin{bmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{bmatrix} + \begin{bmatrix} cy_1 \\ cy_2 \\ \vdots \\ cy_n \end{bmatrix} = c\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + c\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = c\vec{x}+c\vec{y}. \end{aligned}$$

Linear Combinations

Definition 6: Linear Combinations

A vector $\vec{w}$ is said to be a **linear combination** of the vectors $\vec{v_1}, \vec{v_2}, \ldots, \vec{v_k}$ if there exist scalars $c_1, c_2, \ldots, c_k$ such that

$$\vec{w} = c_1\vec{v_1} + c_2\vec{v_2} + \cdots + c_k\vec{v_k}.$$

The scalars c_i are called the **coefficients** of the linear combination. In the case when $k=1$, $\vec{w} = c_1\vec{v_1}$ is said to be **parallel** to $\vec{v_1}$.

Example 7. The following are linear combinations of $\vec{v_1}, \vec{v_2}$, and $\vec{v_3}$:

- $2\vec{v_1} - 3\vec{v_2} + \vec{v_3}$
- $-4\vec{v_1} - 1.0043\vec{v_2} + \sqrt{2}\vec{v_3}$

- $\vec{0} = 0\vec{v_1} + 0\vec{v_2} + 0\vec{v_3}$

Example 8. Let $\vec{v_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\vec{v_2} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Determine which of the following vectors is a linear combination of $\vec{v_1}$ and $\vec{v_2}$.

1. $\vec{w_1} = \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}$
2. $\vec{w_2} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$

Solution.

1. We find if there exist scalars c_1 and c_2 such that $\vec{w_1} = c_1\vec{v_1} + c_2\vec{v_2}$. This equation yields

$$\begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Simplifying the right hand side of the equation gives

$$\begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} = \begin{bmatrix} c_1 + c_2 \\ c_1 \\ c_2 \end{bmatrix},$$

from which it follows that $c_1 = 2$ and $c_2 = -3$, by comparing components. Therefore,

$$\vec{w_1} = 2\vec{v_1} - 3\vec{v_2}$$

and that $\vec{w_1}$ is a linear combination of $\vec{v_1}$ and $\vec{v_2}$.

2. We check if there exist scalars c_1 and c_2 such that $\vec{w_2} = c_1\vec{v_1} + c_2\vec{v_2}$. This equation gives

$$\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Simplifying the right hand side of the equation yields

$$\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} c_1 + c_2 \\ c_1 \\ c_2 \end{bmatrix},$$

from which it follows that

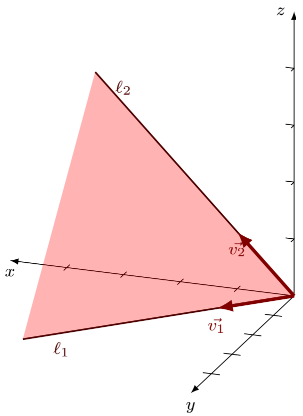
$$\begin{cases} c_1 + c_2 = 2 \\ c_1 \qquad = 1 \\ \qquad c_2 = -1 \end{cases}$$

The second and third equation gives $c_1 = 1$ and $c_2 = -1$, from which it follows that $c_1 + c_2 = 1 + (-1) = 0$ contradicting the first equation. Therefore, $\vec{w_2}$ is **not** a linear combination of $\vec{v_1}$ and $\vec{v_2}$.

In the previous example, if the vectors

$$\vec{v_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \text{ and } \vec{v_2} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

in $\mathbb{R}^3$ are represented in standard positions, the set of all linear combinations of $\vec{v_1}$ and $\vec{v_2}$ consists of all vectors in standard position that lie on the plane through the origin containing the representations of $\vec{v_1}$ and $\vec{v_2}$.



We can argue that every vector of the form $c_1\vec{v_1}$ lies in the line ℓ_1 through the origin that contains $\vec{v_1}$ and that every vector of the form $c_2\vec{v_2}$ lies in the line ℓ_2 through the origin that contains the vector $\vec{v_2}$. Hence, by the parallelogram rule, the linear combination $c_1\vec{v_1} + c_2\vec{v_2}$ must lie in the same plane as the lines ℓ_1 and ℓ_2.

Example 9. Let $\vec{e_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{e_2} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\vec{e_3} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. If $\vec{v} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$ is any vector in $\mathbb{R}^3$, then

$$\vec{v} = \begin{bmatrix} c_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ c_2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ c_3 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Thus, $\vec{v} = c_1\vec{v_1} + c_2\vec{v_2} + c_3\vec{v_3}$. Therefore, every vector in $\mathbb{R}^3$ is a linear combination of the vectors $\vec{e_1}$, $\vec{e_2}$, and $\vec{e_3}$. The vectors $\vec{e_1}$, $\vec{e_2}$, and $\vec{e_3}$ are called the **standard unit vectors** in $\mathbb{R}^3$.

Example 10. Determine value(s) of t such that $\begin{bmatrix} 1 \\ t^2 \end{bmatrix}$ is parallel to $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$.

Solution. The given vectors are parallel if and only if there exists a scalar k such that

$$k \begin{bmatrix} 1 \\ t^2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \iff \begin{bmatrix} k \\ kt^2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

Looking at the first components, we see that $k = 4$. Now, looking at the second components, we have $4t^2 = 1$. This equation can be solved as follows:

$$4t^2 - 1 = 0 \iff (2t-1)(2t+1) = 0 \iff t = \pm\tfrac{1}{2}.$$

▨

Example 11 (Application: RGB-color system)**.** The RGB-color system is a mathematical modeling of colors, where the *primary colors* are **r**ed, **g**reen, and **b**lue. This system is being used in the production of computer screens including personal computers, mobile phones, tablets, and televisions. In this system, we assign

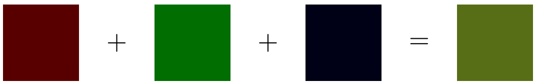
$$\text{red} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \text{green} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ and blue} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

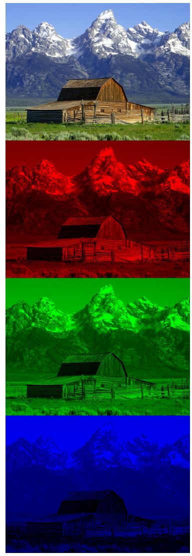
A **color** in the spectrum is considered as any linear combination of the form

$$C = c_1 \cdot \text{red} + c_2 \cdot \text{green} + c_3 \cdot \text{blue} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix},$$

where $0 \leq c_i \leq 1$ for $i = 1, 2, 3$.

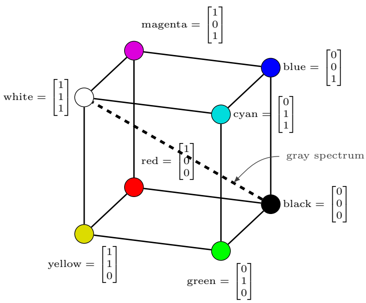
As an example, the color $C = \begin{bmatrix} 0.4 \\ 0.5 \\ 0.1 \end{bmatrix}$ is the color which is 40% red, 50% green and 10% blue.





Source: Wikipedia

In this color model, the color **white** is defined as the vector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \text{red} + \text{green} + \text{blue}$, which is the combination of full red, full green, and full blue. In contrast, the zero vector represents the color **black** (absence of the primary colors).



The diagram on the left shows how the spectrum of colors may be represented by the points in the unit cube of $\mathbb{R}^3$, also known as the **RGB-color space**.